

MATH EXAM-01 — DETAILED MARKING REPORT

Units 1, 2 — Real Numbers, Logarithms

Seemab · Class 9 · Attempt Date: 2026-04-18

 **PERFECT SCORE: 65 / 65 · 100% · Grade A+**

First flawless exam — every question correct, every mark earned

12 / 12

Section A (MCQs)
100%

33 / 33

Section B (Short Qs)
100%

20 / 20

Section C (Long Qs)
100%

SECTION A — Multiple Choice Questions (12 / 12)

#	Question (summary)	Correct	Seemab's	Result
1	Which is an irrational number?	C ($\sqrt{5}$)	C	1
2	$a(b+c) = ab+ac$ is:	C (Distributive)	C	1
3	$\mathbb{R}$ is the union of:	C (Rationals & Irrationals)	C	1
4	Additive inverse of -7	D (7)	D	1
5	In $\sqrt[3]{25}$, the number 25 is called:	C (Radicand)	C	1
6	a^0 ($a \neq 0$) equals:	B (1)	B	1
7	Scientific notation of 0.000052	A (5.2×10^{-5})	A	1
8	$\log_2 64 = y$	C (6)	C	1
9	Characteristic of $\log 456.7$	C (2)	C	1
10	$\log_b(m \times n)$ equals:	A ($\log m + \log n$)	A	1
11	Base of natural logarithm	D (e)	D	1
12	$\log_b 1$ equals	A (0)	A	1

SECTION B — Short Questions (33 / 33)

Q.2(i) — Five properties of real numbers (addition) 3 / 3

Seemab wrote: Closure ($\forall a, b \in \mathbb{R}, a+b \in \mathbb{R}$), Associative ($(a+b)+c = a+(b+c)$), Commutative ($a+b = b+a$), Additive Identity ($a+0 = a$), Additive Inverse ($a + (-a) = 0$). All five with correct general forms.

Complete. Every property named and stated with correct algebraic form.

Q.2(ii) — Identify the property used 3 / 3

Seemab wrote: (a) $5 \times 7 = 7 \times 5 \rightarrow$ Commutative $\checkmark$ (b) $3 + (4+2) = (3+4) + 2 \rightarrow$ Associative $\checkmark$ (c) $8 \times 1 = 8 \rightarrow$ Multiplicative Identity $\checkmark$

All three correctly identified.

Q.2(iii) OR — Single radical and simplify 3 / 3

Seemab wrote: (a) $\sqrt[3]{6} \cdot \sqrt[3]{5} = \sqrt[3]{6 \times 5} = \sqrt[3]{30}$ $\checkmark$ (b) $\sqrt{20} \div \sqrt{5} = \sqrt{(20/5)} = \sqrt{4} = 2$ $\checkmark$

Clean application of product and quotient rules for radicals.

Q.2(iv) OR — Write $x^{3/4}$ as a radical 3 / 3

Seemab wrote: $x^{3/4} = (x^3)^{1/4} = \sqrt[4]{x^3}$ $\checkmark$

Exact application of $b^{m/n} = \sqrt[n]{b^m}$.

Q.2(v) — Solve $\sqrt{3x+1} = 5$ 3 / 3

Seemab wrote: Squared both sides: $3x+1=25 \rightarrow 3x=24 \rightarrow \mathbf{x=8}$. Then verified: $\sqrt{3 \cdot 8 + 1} = \sqrt{25} = 5 \checkmark$. Solution set $\{8\}$.

Shows squaring, isolating, and verification — exactly the right procedure for a radical equation.

Q.2(vi) — Scientific notation 3 / 3

Seemab wrote: (a) $7,340,000 = 7.34 \times 10^6 \checkmark$ (b) $0.000089 = 8.9 \times 10^{-5} \checkmark$ (c) $125.6 = 1.256 \times 10^2 \checkmark$

All three conversions correct including the negative exponent case.

Q.2(vii) OR — Find x in log equations 3 / 3

Seemab wrote: (a) $\log_3 x = 4 \rightarrow x = 3^4 = \mathbf{81} \checkmark$ (b) $\log_x 16 = 2 \rightarrow x^2 = 16 \rightarrow \mathbf{x=4} \checkmark$ (c) $\log_4 64 = x \rightarrow 4^x = 64 = 4^3 \rightarrow \mathbf{x=3} \checkmark$

Confident use of $\log_b a = c \implies b^c = a$ on all three variations.

Q.2(viii) OR — Common vs Natural logarithm 3 / 3

Seemab wrote: 3 differences in tabular form — Base (10 rational vs e irrational), Notation ($\log_{10} x$ vs $\log_e x$), Made by (Henry Briggs vs John Napier).

Three differences clearly stated, all accurate and textbook-aligned.

Q.2(ix) OR — Characteristic of log 3 / 3

Seemab wrote: (a) $67.3 = 6.73 \times 10^1 \rightarrow \text{char} = \mathbf{1} \checkmark$ (b) $0.00892 = 8.92 \times 10^{-3} \rightarrow \text{char} = \mathbf{3 \text{ } (-3)} \checkmark$ (c) $5 = 5.0 \times 10^0 \rightarrow \text{char} = \mathbf{0} \checkmark$

Correctly used bar-notation for negative characteristic — good conventional practice.

Q.2(x) OR — Combine into single log 3 / 3

Seemab wrote: $2 \log x - 3 \log y + \log z = \log x^2 - \log y^3 + \log z = \mathbf{\log(x^2z/y^3)} \checkmark$

Applied power law then quotient/product laws in correct sequence.

Q.2(xi) — Number of digits in 2^{20} ($\log 2 = 0.3010$) 3 / 3

Seemab wrote: $\log(2^{20}) = 20 \times 0.3010 = 6.02$. Characteristic = 6. Digits = char + 1 = **7 digits** $\checkmark$

Applied the "digits = characteristic + 1" rule correctly after the log computation.

SECTION C — Long Questions (20 / 20)

Q.3(i) OR — Prove $(x^p/x^q)^{p+q} \cdot (x^q/x^r)^{q+r} \cdot (x^r/x^p)^{r+p} = 1$ 5 / 5

Seemab wrote: Used quotient rule then power rule:

$$\begin{aligned} &= (x^{p-q})^{p+q} \cdot (x^{q-r})^{q+r} \cdot (x^{r-p})^{r+p} \\ &= x^{p^2-q^2} \cdot x^{q^2-r^2} \cdot x^{r^2-p^2} \\ &= x^{(p^2-q^2)+(q^2-r^2)+(r^2-p^2)} = x^0 = \mathbf{1} \checkmark \end{aligned}$$

Elegant use of the difference-of-squares identity $(a-b)(a+b) = a^2-b^2$. Each exponent pair cancels beautifully. Full proof.

Q.3(ii) — Number line plots + identify properties 5 / 5

Seemab wrote: Number line plots for (a) -3 , (b) $7/4 = 1.75$, (c) $\sqrt{5} \approx 2.236$, (d) $-2 < x < 3$ with open circles — all drawn accurately. Properties: (ii) $(x+y)+z = x+(y+z) \rightarrow$ **Associative** $\checkmark$ (iii) $5 \cdot (1/5) = 1 \rightarrow$ **Multiplicative Inverse** $\checkmark$

Four number-line plots and two property identifications, all correct. Good attention to open vs closed circles on the inequality.

Q.3(iii) — Prove product law + evaluate $\log 72$ 5 / 5

Proof: Let $\log_b m = x$, $\log_b n = y$. So $m = b^x$, $n = b^y$. $mn = b^x \cdot b^y = b^{x+y}$. Therefore $\log_b(mn) = x+y = \log_b m + \log_b n \checkmark$
log 72: $72 = 2^3 \cdot 3^2$. $\log 72 = 3 \log 2 + 2 \log 3 = 3(0.3010) + 2(0.4771) = 0.9030 + 0.9542 = \mathbf{1.8572} \checkmark$

Rigorous proof using the definition of logarithm and the exponent product rule. Numerical evaluation carried through with textbook values.

Q.3(iv) OR — Digits in 5^{37} ($\log 5 = 0.6990$) 5 / 5

Seemab wrote: $\log(5^{37}) = 37 \times 0.6990 = 25.863$. Characteristic = 25. Digits = characteristic + 1 = **26 digits** ✓

Clean application of the same "digits = characteristic + 1" rule she used in Q.2(xi), now with a harder base.

What Made This a Perfect Score

- **Every sub-part answered.** No blank spaces, no skipped questions, no rushed endings. Unlike the last exam, every single letter sub-part of every question got an answer.
- **Verification shown where relevant.** Q.2(v) solved $\sqrt{(3x+1)}=5$, got $x=8$, *then checked* by substituting back and confirming $\sqrt{25}=5$. That's textbook discipline.
- **Correct notation for negative characteristic.** Q.2(ix)(b) used the bar notation 3 for -3 — the convention Pakistani curricula teach explicitly.
- **Clean algebra on Q.3(i).** The $(x^p/x^q)^{(p+q)}$ identity can trap students who forget $(a-b)(a+b)=a^2-b^2$. Seemab used it cleanly and let the exponents cancel.
- **Tabular structure for comparisons.** Q.2(viii) Common vs Natural — clean table, 3 valid differences.
- **Correctly solved for x in $\log_x 16 = 2$.** Many students get confused when the variable is the base; Seemab inverted cleanly to get $x=4$.

Overall: Math Units 1-2 material is clearly mastered — Real Numbers (properties, radicals, scientific notation) and Logarithms (definition, laws, characteristic, digit-count applications). This is the kind of paper that shows both conceptual depth and exam-technique discipline. Building on the lessons learned from Chem exam-02 (finishing every sub-part, double-checking arithmetic), Seemab has translated those improvements into her first perfect score.